

RISHABH RASTOGI

Ph.D. · Technical Advisor & Operator · Semiconductor | Deep Tech | Investment Intelligence

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PROFILE

Semiconductor technologist and independent advisor with a rare combination of atomic-scale process expertise and the ability to translate it into business and investment decisions. Built on 8+ years of hands-on fabrication at IMEC's 300mm fab (world's leading semiconductor R&D consortium) and ASM International, I bring the depth of a process engineer to the boardroom, not the cleanroom. I help PE firms, hedge funds, deep tech startups, and equipment companies make faster, smarter decisions — on technology bets, competitive positioning, equipment evaluation, and market entry — because I have actually built what they are evaluating. 7 patents, 6 publications, 5 countries. Currently available for senior advisory, business development, technical strategy, and startup operator roles globally — with a preference for India-based or India-facing roles.

WHAT I BRING

Technical credibility at the frontier: IMEC pedigree, 7 patents, sub-3nm GAA process development — the deepest possible domain authority in semiconductor manufacturing.

Translation as a commercial instrument: Proven ability to make invisible atomic-scale complexity legible to investors, executives, and business stakeholders — enabling decisions, not just information transfer.

Operator range: Co-founded two ventures, managed enterprise client relationships, built cross-continental workflow systems. Not a pure technical specialist — a technically credentialed operator.

Global range, India-rooted: Worked across Belgium, Luxembourg, France, USA, and India. Culturally fluent across multinational teams. Currently building deep tech and advisory presence in the Indian market.

PROFESSIONAL EXPERIENCE

Independent Consultant — Semiconductor & Deep Tech Advisory | Self-employed

Mar 2025 – Present • Remote, India

Technical due diligence and strategic advisory for PE firms, hedge funds, and deep tech startups — operating via Tegus by AlphaSense expert network and direct client engagements.

- Deliver expert advisory engagements across multiple institutional investment networks — covering High-NA EUV forecasting, advanced lithography market dynamics, ASM International competitive positioning, TEL market trajectory, and China trade restriction impact on the WFE supply chain — enabling PE firms, hedge funds, and asset managers to connect process-level technical realities to capital allocation decisions.
- Served as part-time technical advisor to a Switzerland-based semiconductor startup on establishing a 28nm logic fab from ground up — scoping process architecture, equipment requirements, and resource planning; engagement concluded due to client-side funding constraints.
- Conduct ALD/CVD process feasibility assessments and equipment benchmarking for deep tech and OEM clients — evaluating integration strategies for advanced logic and memory device flows at sub-3nm and legacy nodes.
- Deliver guest lectures and workshops for undergraduate and early-career audiences at universities in India and abroad — translating semiconductor manufacturing realities and deep tech career pathways into accessible, engaging formats for the next generation of scientists and engineers.
- Consistently recognised for ability to bridge atomic-scale technical knowledge and investment-relevant narrative — enabling clients to connect technology development phases to market timelines and capital allocation decisions.

Senior Process Engineer → Customer-Facing Technical Lead | ASM International N.V.

Jun 2022 – Mar 2025 • USA / Belgium — embedded at IMEC 300mm fab, Leuven

Led ALD/CVD process development for sub-3nm GAA logic nodes at IMEC's 300mm research foundry — the world's most advanced semiconductor R&D environment — while serving as the primary technical interface for Tier-1 customer engineering teams.

- **Commercial impact:** Served as primary technical communicator for customer-facing discussions with Tier-1 fabs — consistently recognised for clarity, confidence, and ability to articulate complex process trade-offs in language accessible to customer engineers and decision-makers.
- **Technical delivery:** Developed 15+ ALD/CVD deposition recipes for sub-3nm GAA integration — gate stack dielectrics, rare earth oxide dipole materials (LaO, YO), high-k interfacial layers — contributing to 7 patent filings and establishing ASM's process position at the leading edge.
- **Cross-functional execution:** Managed 4+ process transfers from alpha development through HVM beta qualification at customer fabs (TSMC, Intel, Micron) — coordinating documentation, on-site support, and structured troubleshooting across US and Belgian teams.
- **Operational systems:** Built cross-continental workflow and communication tools adopted by teams across geographies — improving process transparency, accountability, and knowledge transfer without formal mandate.
- **Knowledge transfer:** Delivered knowledge-sharing seminars across US and Belgium R&D teams; represented ASM in direct customer technical meetings with recognised impact on relationship quality and customer confidence.

Co-Founder | Flamboyant Technologies & SanWalk

Feb 2021 – May 2022 • India

- Flamboyant Technologies: Business development, client acquisition, and solution scoping for ERP implementations (MS Dynamics 365) across 10+ enterprise clients — first experience owning a commercial pipeline end-to-end.
- SanWalk: Led product concept, go-to-market strategy, MVP design, and early-stage prototyping for a consumer hardware venture — first-principles startup execution without institutional support.

Research Engineer (Doctoral) | Luxembourg Institute of Science and Technology (LIST)

Nov 2016 – Nov 2020 • Luxembourg / France

- Conducted doctoral research in nanoscale thin film engineering and nanoplasmonic device fabrication — published 6 first-author papers, filed 2 patents, delivered invited oral presentations at international conferences including ICES (Canada) and SPIE.
- Developed the scientific and communication foundation that now underpins technical advisory work — translating experimental nanoscience into publishable, peer-reviewed narratives accessible to interdisciplinary audiences.

EDUCATION

Ph.D. — Materials Science and Nanoscale Engineering University of Technology of Troyes, France | 2016–2020

B.Tech + M.Tech — Nanotechnology (Devices) Amity University, Noida, India | 2010–2015

CREDENTIALS & RECOGNITION

- 7 patents filed — sub-3nm GAA gate stack dielectrics, rare earth oxide dipoles, gapfill architectures (ASM International, 2024–2025)
- 6 peer-reviewed publications — ACS Applied Materials & Interfaces, Journal of Physical Chemistry C, Sensors and Actuators
- Expert network consultant — Tegus by AlphaSense (semiconductor process technology, WFE market, advanced node strategy)
- Six Sigma Green/Black Belt · DOE (JMP) · SPC · IQ/OQ/PQ process qualification
- International conference speaker — ICES Canada (invited oral), SPIE, NFO; 7 total presentations
- 5 countries of professional experience: India, Luxembourg, France, Belgium, USA

AREAS OF ENGAGEMENT

Technical Due Diligence · Semiconductor Market Advisory · Equipment & Process Evaluation · Business Development (Deep Tech) · Technology Strategy & Roadmapping · Startup Technical Co-founder / Operator · R&D to Commercialisation · Investor Education (Advanced Node Technologies)

INDEPENDENT PROJECTS & EMERGING SKILLS

- Built and deployed multiple live web platforms independently using VS Code, GitHub, and Netlify — including a professional portfolio site and sector-specific advisory/advocacy platforms — with zero external engineering support, using AI-augmented development workflows (Claude, ChatGPT).
- Designed and deployed a personal equity and futures market tracker dashboard — integrating live price feeds, watchlist management, and TradingView links — as a self-built tool supporting active trading and investment management across US equities, gold futures (MCX), and Indian markets.
- Active retail trader and self-directed investor across US equities, futures, and Indian capital markets — bringing first-hand investor perspective to financial sector advisory engagements.
- Developing personal brand presence on LinkedIn and Instagram — creating original content on semiconductor technology, deep tech investing, and advanced node R&D using AI-assisted content workflows; building audience in the intersection of technical and investor communities.

Full patents and publications list available on request.